INCH-POUND

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27 November 2023
SUPERSEDING
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DETAIL SPECIFICATION

AMINOGUANIDINE BICARBONATE (AMINATE)

Reinstated after 27 November 2023 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests on aminoguanidine bicarbonate for use in the manufacture of tetracene.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1193	-	Standard Specification for Reagent Water
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Material. The aminoguanidine bicarbonate shall be a white or cream powder.

3.3 Chemical properties. The aminoguanidine bicarbonate shall conform to the chemical requirements shown in Table I.

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TABLE I. Chemical requirements for aminoguanidine bicarbonate.

Properties	Requirements
Aminoguanidine bicarbonate assay Weight percent minimum, (wt% min). Weight percent maximum, (wt% max).	 98.0 100.8
Acid-insoluble residue, wt% max.	0.02
Moisture, wt% max.	0.30
Ash, wt% max.	0.10

3.4 Workmanship. The aminoguanidine bicarbonate shall be uniform in quality, free from visible impurities and foreign material such as dirt, wooden particles, or any other foreign matter.

4. VERIFICATION

TABLE II. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Material			X		X	X	4.4.6
3.3	Aminoguanidine bicarbonate assay	X			X	X	X	4.4.2
3.3	Acid-insoluble residue	X			X	X	X	4.4.3
3.3	Moisture	X			X	X	X	4.4.4
3.3	Ash	X			X	X	X	4.4.5
3.4	Workmanship			X	X	X	X	4.4.6

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

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4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table III. Samples of not less than 1 pound (lb.) of aminoguanidine bicarbonate collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

TABLE III. First article acceptance tests.

Category	Examination or test	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	Material	3.2	4.4.6
102	Aminoguanidine bicarbonate assay	3.3	4.4.2
103	Acid-insoluble residue	3.3	4.4.3
104	Moisture	3.3	4.4.4
105	Ash	3.3	4.4.5
Minor			
201	Workmanship	3.4	4.4.6

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements, the first article batch shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." Each batch shall consist of a quantity of aminoguanidine bicarbonate that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one acceptable batch of aminoguanidine bicarbonate produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2. Aminoguanidine sampling. Sampling for inspection shall be performed at either the supplier/distributor location where the aminoguanidine is packed, or at the manufacturer's receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Sampling at the manufacturer. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 1 lb. shall be formed which is assumed to be representative from a batch. All composite samples shall be verified in accordance with 4.3.3. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.2.2 Sampling at the receiving facility. Aminoguanidine batches packaged in shipping containers (e.g., drums, totes, boxes, etc.) shall be sampled and accepted using

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Verification Level II of MIL-STD-1916, where one unit is defined as one container. Each container shall be sampled in three (3) locations of the shipping containers in accordance with ASTM E300 using a thief stick. The samples from each shipping container shall be examined and be placed in a secondary airtight container. The samples shall be mixed to form a composite sample consisting of a minimum of 1 lb. for each aminoguanidine bicarbonate lot received. There shall be no reduced sampling. Every lot received shall be tested in accordance with 4.3.3. If any sample fails to comply with any applicable test requirements the lot represented by the sample shall be rejected.

4.3.3 Aminoguanidine bicarbonate.				
Category	Examination or test	Verification level	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Material	4.3.2	3.2	4.4.6
102	Assay	4.3.2	3.3	4.4.2
103	Acid-insoluble residue	4.3.2	3.3	4.4.3
104	Moisture	4.3.2	3.3	4.4.4
105	Ash	4.3.2	3.3	4.4.5
Minor				
201	Workmanship	4.3.2	3.4	4.4.6

4.4 Tests.

4.4.1 Preparation of reagents. Unless otherwise specified in the contract, all chemicals and reagents shall be reagent grade or better. Water shall be distilled, deionized, or equivalent, conforming to ASTM D1193 Type I or Type II. Standard 0.1N sodium hydroxide solution may be purchased commercially or standardized against potassium hydrogen phthalate using phenolphthalein indicator.

4.4.1.1 Ferrocyphen indicator solution. Dissolve 0.1 grams (g) of “ferrocyphen” in 10 milliliters (mL) of concentrated sulfuric acid. “Ferrocyphen” is dicyano-bis-(1,10 phenanthroline)-iron II dihydrate and can be prepared as follows: Add 3 g of 1,10 phenanthroline monohydrate and 1.95 g of $FeSO_4 \cdot 7\ H_2O$ to 200 mL of water and heat just below the boiling point. Add at one stroke 5 g of potassium cyanide dissolved in 5 mL of water. Stir occasionally for a few minutes on the hot plate and then allow to stand overnight at room temperature. Filter through a Buchner funnel, wash well with water, spread out on a watch glass, and allow to dry overnight at room temperature.

4.4.1.2 Sulfanilic acid. Fresh sulfanilic acid shall be used (see 6.3.2). Immediately prior to use, sulfanilic acid shall be pulverized with a mortar and pestle, mixed well, and placed in a glass-stoppered bottle. Determine the purity by dissolving 0.5 g (weighed to 0.1 milligrams) in 100 mL of water and titrating with standard 0.1N sodium hydroxide solution to a phenolphthalein or bromothymol blue end point. Calculate the purity as follows:

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$$\text{Percent sulfanilic acid (anhydrous)} = \frac{17.32 \times \text{mL of NaOH} \times \text{normality of NaOH}}{\text{grams of sample}}$$

4.4.1.3 Standard sodium nitrite solution (0.5N). Dissolve 34.5 g of reagent grade sodium nitrite ($NaNO_2$) in water and dilute to 1 liter (L). To standardize, transfer 2 g of the sulfanilic acid (weighed to 0.1 milligrams) to a beaker of about 400 mL in size. Dissolve in a mixture of 150 mL of water and 100 mL of hydrochloric acid by stirring with a stirring rod. Add 1 mL of "ferrocyphen" indicator solution and proceed with the titration with sodium nitrite solution as described under 4.4.2. Calculate the normality of the sodium nitrite solution as follows:

$$\text{Normality of } NaNO_2 = \frac{0.05774 \times \text{grams of sulfanilic acid} \times \text{purity of sulfanilic acid in percentage (\%)}}{\text{mL. of } NaNO_2}$$

4.4.2 Aminoguanidine bicarbonate assay. Transfer 1.7 g of the sample (weighed to 0.1 milligrams) to a beaker of about 400 mL in size, add 150 mL of water, and start magnetic stirring. Add 100 mL of hydrochloric acid (the first 5-10 mL slowly). Add 1 mL of "ferrocyphen" indicator solution. Insert the tip of a 50 ml class A burette below the surface of the solution and titrate with the standard sodium nitrite solution. Titrate fairly rapidly at first and dropwise near the end point. Allow 10-15 seconds between increments near the end point. The end point is a change from faint orange to a pure pale violet when viewed laterally through the vortex of the rapidly stirred solution against a white background. Calculate the percentage of aminoguanidine bicarbonate as follows:

$$\text{Percentage aminoguanidine bicarbonate} = \frac{13.61 \times \text{mL of } NaNO_2 \times \text{normality of } NaNO_2}{\text{grams of sample}}$$

4.4.3 Acid-insoluble residue. Dissolve a 10 g sample in a beaker of about 250 mL in size with 50 mL of water and 15 mL of hydrochloric acid. Filter through a tared (weighed to the nearest 0.1 mg) sintered glass crucible of medium porosity, transfer, and wash with water. Dry at 110 ºC for 1 hour, cool in a desiccator, and weigh to the nearest 0.1 mg. Calculate as follows:

$$\text{Percentage insoluble matter} = \frac{\text{grams of residue}}{\text{grams of sample}} \times 100$$

4.4.4 Moisture. Transfer a 10 g sample (weighed to 1.0 mg) to an open weighing dish of about 6 centimeters in diameter, measure the gross weight to the nearest 0.1 mg, place in a desiccator containing fresh sulfuric acid, and allow to stand overnight. Determine the loss in weight. Calculate as follows:

$$\text{Percent moisture} = \frac{\text{grams loss in weight}}{\text{grams of sample}} \times 100$$

4.4.5 Ash. Transfer a 5 g sample to a platinum crucible previously tared to the nearest 0.1 mg. Place the crucible on the hot plate at highest heat until the strong effervescence has ceased. Heat over a Meker burner or in a muffle furnace at 800 °C until only small residue remains. Add 5 mL of nitric acid, evaporate to dryness on the hot plate, and ignite for a few

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minutes over a Meker burner or in a muffle furnace at 800°C. Cool in a desiccator and weigh to the nearest 0.1 mg. Calculate as follows:

$$\text{Percent ash} = \frac{\text{grams of residue}}{\text{grams of sample}} \times 100$$

4.4.6 Workmanship. Take approximately 200 g of the material and spread the composition out on a clean surface. The sample shall be visually examined and shall be uniform in quality, free from lumps, grit, visible impurities, foreign matter, or other defects. Failure to meet the workmanship requirement during inspection or while executing production processes shall result in batch rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Aminoguanidine bicarbonate covered by this specification is intended for use as ingredient in explosives.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Provisions for submission/resubmission of first article samples
c. Quantity required and delivery schedules
d. Level of packaging and packing required (see 5.1).
e. Certificate(s) of Analysis reporting the test results for the batch(es) or container(s) Tested

6.3 Definitions.

6.3.1 Batch. A discrete quantity of material that has been manufactured through various unit operations to produce a substantially uniform final product.

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6.3.2 Fresh sulfanilic acid. Sulfanilic acid may deteriorate on prolonged storage. It is strongly recommended to obtain fresh sulfanilic acid and store it in accordance with manufacturer recommendation.

6.4 Technical agency. US Army Combat Capability Development Command Armaments Center (DEVCOM AC) is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods and/or instruments. Prior approval by the Contracting Officer should be required for use of equivalent test methods and/or instruments. A description of the proposed method and/or instrumentation should be submitted through the Contracting Officer to the Technical Agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.4) responsible for preparation of this specification.

6.7 Incorporated changes. ECP 23Q2004 provides documentation for noted changes.

6.8 Subject term (key word) listing.

Ingredient
Tetracene

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Army – AR
Navy – OS
Air Force – 70

Preparing activity:
Army – AR
(Project 6810-2023-005)

Review Activities:
Army – AV, EA, MI, MR, TE
Navy – AS
Air Force – 11, 68

DLA - GS

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